Lec-2

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$l=2$
$r=4$

$(a_1\ a_2\ a_3\ a_4)\ (b_1\ b_2\ b_3\ b_4)$

$f(l,r) = \Sigma a_i b_i$, $1 \le i \le 2$ | $f(2,4) = \Sigma a_i b_i$, $2 \le i \le 4$ | $= a_2b_2 + a_3b_3 + a_4b_4$

You are given two arrays a and b, both of length n.

Let's define a function $f(l,r) = \sum_{l \le i \le r} a_i \cdot b_i$.

Your task is to reorder the elements (choose an arbitrary order of elements) of the array b to minimize the value of $\sum_{1 \le l \le r \le n} f(l,r)$. Since the answer can be very large, you have to print it modulo 998244353. Note that you should **minimize the answer but not its remainder**.

Input

The first line of the input contains one integer n ($1 \le n \le 2 \cdot 10^5$) — the number of elements in a and b.

The second line of the input contains n integers $a_1, a_2, \ldots, a_n$ ($1 \le a_i \le 10^6$), where a_i is the i-th element of a.

The third line of the input contains n integers $b_1, b_2, \ldots, b_n$ ($1 \le b_j \le 10^6$), where b_j is the j-th element of b.

Output

Print one integer — the minimum possible value of $\sum_{1 \le l \le r \le n} f(l,r)$ after rearranging elements of b, taken modulo 998244353. Note that you should **minimize the answer but not its remainder**.

0 1 2 3
$(a_1\ a_2\ a_3\ a_4)$

0 1 2 3
$(b_1\ b_2\ b_3\ b_4)$

0-0
0-1
0-2
0-3

1-1
1-2
1-3

2-2
2-3

2-3

$f(1-1) = \Sigma a_i b_i = a_2b_2$
$f(1-2) = \Sigma a_i b_i = a_2b_2 + a_3b_3$
$f(1-3) = \Sigma a_i b_i = a_2b_2 + a_3b_3 + a_4b_4$
$f(2-2) = \Sigma a_i b_i = a_3b_3$
$f(2-3) = \Sigma a_i b_i = a_3b_3 + a_2b_4$
$f(3-3) = \Sigma a_i b_i = a_4b_4$

$f(0,0) = \Sigma a_i b_i = a_1b_1$
$f(0-1) = \Sigma a_i b_i = a_1b_1 + a_2b_2$
$f(0-2) = \Sigma a_i b_i = a_1b_1 + a_2b_2 + a_3b_3$
$f(0-3) = \Sigma (a_i b_i) = a_1b_1 + a_2b_2 + a_3b_3 + a_4b_4$

1 2 3
5 7 6

0-0
0-1
0-2

1-1
1-2

2-2

2
(1 3)
(4 2)

0-0
0-1

1-1

0-0 → 4
0-1 → 1×4 + 3×2 = 10
1-1 → 3×2 = 6

20

1 3
2 4

0-0 → 1×2 = 2
0-1 → 1×2 + 3×4 = 14
1-1 → 3×4 = 12
28

| 9 | 81 | 130 |

5
18724
97293

(1, 8, 7, 2, 4)
(9, 9, 7, 2, 3)

0-0 → 9
0-1 → 9+72 = 81
0-2 → 9+72+49 = 130
0-3
0-4

1-1
1-2
1-3
1-4

2-2
2-3
2-4

3-3
3-4

4-4

$l \to i+1$
$r \to n-i$ → 4

a_1 a_2 a_3 a_4 a_5 a_6

0 1 2
1
3

$a_1 \to l = 1$, $r = 6$

$a_2 = l \to 2$, $r = 5$

$a_3 = l = 3$, $r = 3$

$nc_1 = \frac{n!}{(n-1)!\,1!}$

$= \frac{n!}{(n-1)!\,1!}$

$= \frac{n \times \cancel{(n-1)!}}{\cancel{(n-1)!} \times 1}$

a_1 a_2 a_3 a_4 a_5 a_6

$l = i+1 \to i+1c_1$
$r = n-i \to n-ic_1$

$b_1\ ([a_3]\ (i+1)\ (n-i))$

$(b_2\ b_6\ b_1\ b_3\ b_2\ b_5)$

4-3 = 3
4-2 = 2
4-3 = 0

0 1 2 3
$(a_1\ a_2\ a_3\ a_4)$

$(b_1\ b_2\ b_3\ b_4)$

$a_1 = 1 \times 4 = 4$

$a_2 = 2 \times 3 = 6$

$a_3 = 3 \times 2 = 6$

$a_4 = 4 \times 1$

$4a_1\ b_4$ | $6a_2\ b_1$ | $6a_3\ b_2$ | $4a_4\ b_3$

$4a_1b_4 + 6a_2$